



Kardar-Parisi-Zhang Scaling in Time-Crystalline Matter

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Forming a continuous-time crystal leads to **spontaneous symmetry breaking of time-translation invariance**, generating a **massless Goldstone mode**.

Question: What is the Goldstone mode? what properties it has?

Answer: The Goldstone mode is an angular variable which belongs to **KPZ universal class**.

continuous time crystal(CTC)

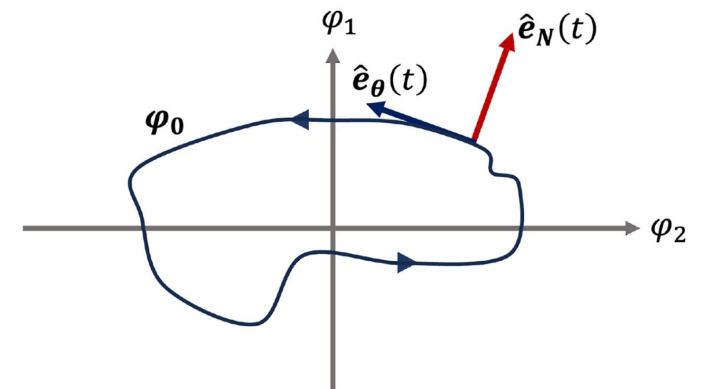
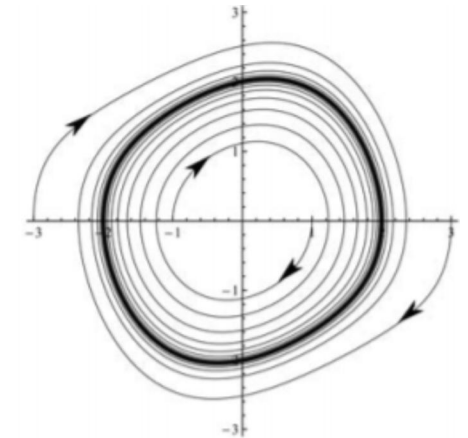
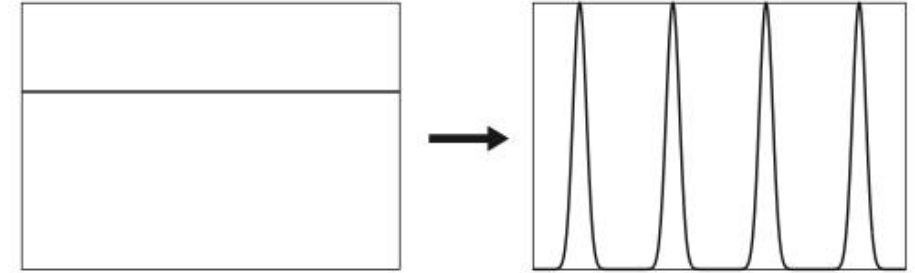
- **definition:** coarse-grained order parameter $\langle \phi(t, \mathbf{x}) \rangle$
is time-periodic with minimal period T
perspective in order parameter phase space: limit cycle

CTC = limit cycle

- **no-go theorem:** CTC cannot exist in equilibrium, can only in non-equilibrium

perspective in order parameter phase space $\varphi(t) = (\varphi_1(t), \varphi_2(t), \dots)$

equilibrium state ($\frac{d\varphi}{dt} = 0$) : fixed point in phase space, not limit cycle



goldstone mode(chronon)

group theory perspective:

- time-translation invariance spontaneous symmetry break from continuous to discrete $\mathbb{R} \rightarrow \mathbb{Z}$
- goldstone mode's symmetry: $\mathbb{R}/\mathbb{Z} = U(1), U(1) = \{e^{i\theta}, \theta \in \mathbb{R}\}$

prove:

define map

$$p : \mathbb{R} \rightarrow U(1), p(x) = e^{2\pi i x}$$
$$\ker(p) = x : e^{2\pi i x} = 1 = \mathbb{Z}$$

so $\mathbb{R}/\mathbb{Z} = U(1)$

goldstone mode is an angular variable

goldstone mode(chronon)

phase space perspective: $\varphi(t) = (\varphi_1(t), \varphi_2(t), \dots)$

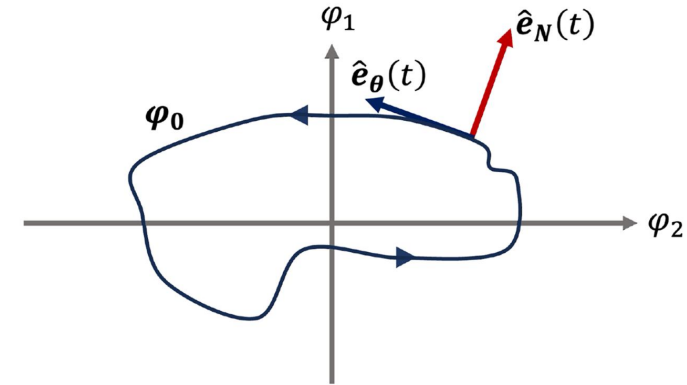
$\langle \phi(t) \rangle$ traces out a one-dimensional closed orbit with period T , which can be parametrized by an angular variable $\theta_0 \in [0, 2\pi)$, $\varphi(t + \theta_0 \bar{T})$ ($\bar{T} = (T/2\pi)$)

fluctuation along $\hat{e}_\theta$ is **gapless** (goldstone mode), perpendicular is gapped

$\theta_0 \rightarrow \theta(t, \mathbf{x})$ which fluctuates on timescales much larger than T

dynamic expression: $\phi(t, \mathbf{x}) = \varphi(t + \bar{T}\theta(t, \mathbf{x}), \mathbf{x}) + N(t, \mathbf{x})$

goldstone mode “chronon”



effective field theory for chronon

kpz

$$\partial_t \theta = \nu \nabla^2 \theta + \frac{\lambda}{2} (\nabla \theta)^2 + \eta(\mathbf{x}, t)$$

- $\frac{\lambda}{2} (\nabla \theta)^2$ break detailed balance
- scaling $x \rightarrow b x, \quad t \rightarrow b^z t, \quad \theta \rightarrow b^\chi \theta$

Dimension	Most relevant nonlinearity	Higher-order terms	Universal exponents	Universality class
$d = 1$	$(\nabla \theta)^2$ (relevant)	irrelevant	$\chi = 1/2, z = 3/2$	KPZ
$d = 2$	marginal $\rightarrow$ strong coupling	irrelevant	$\chi \approx 0.39, z \approx 1.61$	KPZ
$d > 2$	irrelevant	irrelevant	$\chi = (2 - d)/2, z = 2$	EW (Gaussian)

effective field theory for chronon

process: write down an action including all terms ,then drop those prohibited/irrelevant terms

- maximum MSRJD action:
$$S[\theta, \tilde{\theta}] = \int_{t,\mathbf{x}} \left[\tilde{\theta} \mathcal{F}(\theta, \partial_t, \nabla) + \mathcal{D}(\theta, \nabla; \tilde{\theta}) \right]$$
$$\mathcal{F} = a_0 + a_1 \partial_t \theta + a_2 \nabla^2 \theta + \frac{g}{2} (\nabla \theta)^2 + b_1 (\partial_t \theta)^2 + b_2 (\nabla^2 \theta)^2 + b_3 (\nabla \theta)^2 \partial_t \theta + \dots$$
$$\mathcal{D} = D_0 \tilde{\theta}^2 + D_1 (\nabla \tilde{\theta})^2 + D_2 (\nabla \theta)^2 \tilde{\theta}^2 + D_3 \tilde{\theta} \partial_t \tilde{\theta} + \dots$$
- drop prohibited/irrelevant terms
 - translation symmetry: keep invariant under $\theta(\mathbf{x}, t) \rightarrow \theta(\mathbf{x}, t) + c$.
forbbiden terms with undifferentiated θ ,like $\tilde{\theta} \theta, \tilde{\theta} \theta^n, f(\theta) \tilde{\theta}^2$.
 - spatical isotropic: rotation symetry
forbidden Anisotropic,forbidden terms like $\partial_x \theta, \partial_y \theta$
 - Markovianity: long time/wave limit,time/space scale far exceeds that of the memory core.
only allow first-order time derivative and white noise: $\partial_t \theta, D \tilde{\theta}^2$
 - broken detailed balance: allow nonlinear terms like $(\nabla \theta)^2$

effective field theory for chronon

- simplified actions

$$S[\theta, \tilde{\theta}] = \int_{t,x} \tilde{\theta} \left[\partial_t \theta - Z \nabla^2 \theta + \frac{g}{2} (\nabla \theta)^2 \right] - D \tilde{\theta}^2$$

$$\begin{aligned} &+ \tilde{\theta} [-c_3 \partial_i \theta \partial_i \nabla^2 \theta - c_{4a} (\nabla^2 \theta)^2 + \dots] \\ &+ [d_{2a} (\nabla \theta)^2 + d_{2b} \nabla^2 \theta + \dots] \tilde{\theta}^2 \end{aligned}$$

high order terms, irrelevant

- final action: **compact kpz**

$$S = \int_{t,x} \tilde{\theta} \left(\partial_t \theta - Z \nabla^2 \theta + \frac{g}{2} (\nabla \theta)^2 \right) - D \tilde{\theta}^2$$

scaling predictions

chronon belong to compact kpz universality class

- scaling exponent:

$$\ln C(t, 0) \sim -A t^{2\beta}$$

$$\ln C(0, \mathbf{x}) \sim -B |\mathbf{x}|^{2\chi}$$

$$d = 1: \beta = 1/3, \chi = 1/2$$

$$d = 2 \quad \beta \approx 0.24 \quad \chi \approx 0.39$$

$$\boldsymbol{\phi}(t, \mathbf{x}) = \sum_n \mathbf{a}_n \cos[n\Omega t + n\theta(t, \mathbf{x}) + \delta_n]$$

$$C(t, \mathbf{x}) = \langle \boldsymbol{\phi}(t, \mathbf{x}) \cdot \boldsymbol{\phi}(0, 0) \rangle$$

$$= \sum_n \mathbf{a}_n^2 \cos(n\Omega t + \delta_n) e^{-\frac{n^2}{2} \langle (\theta(t, \mathbf{x}) - \theta(0, 0))^2 \rangle_c}$$

$$\sim e^{-\frac{1}{2} \langle (\theta(t, \mathbf{x}) - \theta(0, 0))^2 \rangle_c},$$

- topological defects: cause exponential decay of correlation functions beyond a scale

$L_v \sim e^{\varepsilon/\sigma}$, with activation energy of defects ε , and noise strength σ

Chronons in the many-body Van der Pol

- dynamic equation: $(\partial_t^2 + (2\gamma + u\phi^2 - Z_1 \nabla^2)\partial_t + \omega_0^2 - Z_2 \nabla^2)\phi + \xi$

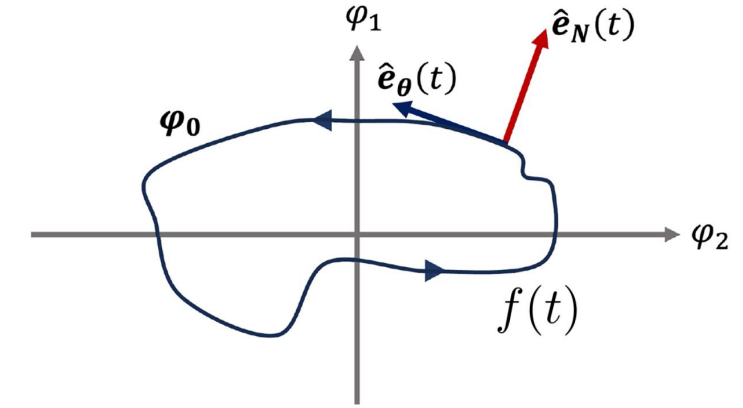
- CTC phase condition: $\gamma < 0$

- phase space: $(\phi_1, \phi_2) = (\phi, \pi = \partial_t \phi)$

- reparameter limit cycle

$$\varphi(t) = (f(t), \dot{f}(t)) \quad \hat{\mathbf{e}}_\theta(t) = \frac{\dot{\varphi}(t)}{\|\dot{\varphi}(t)\|} = \frac{(\dot{f}(t), \ddot{f}(t))^T}{f(t)^2 + \dot{f}(t)^2}, \quad \hat{\mathbf{e}}_N(t) = \frac{(-\ddot{f}(t), \dot{f}(t))^T}{f(t)^2 + \dot{f}(t)^2}.$$

$$\begin{pmatrix} \phi(t, \mathbf{x}) \\ \pi(t, \mathbf{x}) \end{pmatrix} = \begin{pmatrix} f(t + \theta(t, \mathbf{x})\bar{T}) \\ \dot{f}(t + \theta(t, \mathbf{x})\bar{T}) \end{pmatrix} + N(t, \mathbf{x})\hat{\mathbf{e}}_N(t + \theta(t, \mathbf{x})\bar{T})$$



- insert it into dynamic equation

$$\partial_t N - Z_N(t + \theta\bar{T})\nabla^2 N + m_N(t + \theta\bar{T})N + Y_N(t + \theta\bar{T})\nabla^2 \theta + g_N(t + \theta\bar{T})(\nabla \theta)^2 + D_N(t + \theta\bar{T})\xi_N = 0,$$

$$\partial_t \theta - Z_\theta(t + \theta\bar{T})\nabla^2 \theta + Y_\theta(t + \theta\bar{T})\nabla^2 N + m_\theta(t + \theta\bar{T})N + g_\theta(t + \theta\bar{T})(\nabla \theta)^2 + D_\theta(t + \theta\bar{T})\xi_\theta = 0,$$

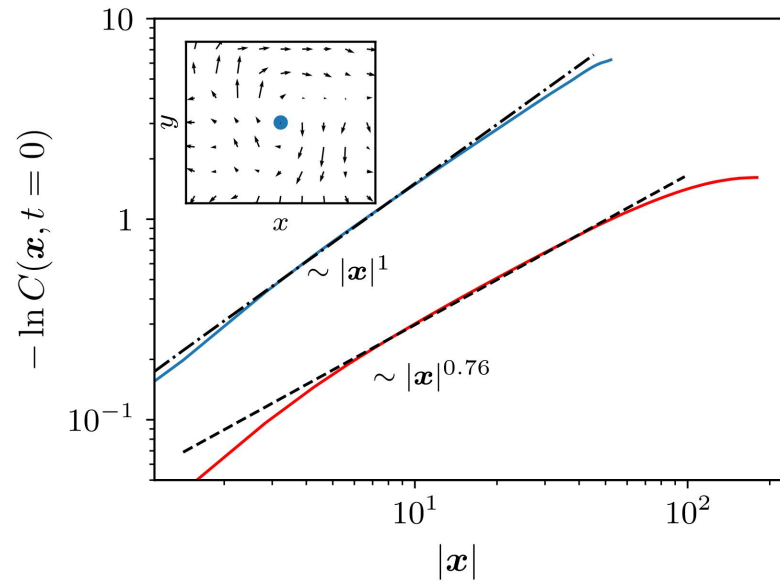
- Integrate out the gapped mode and average over time

$$\partial_t \theta - Z\nabla^2 \theta + \frac{g}{2}(\nabla \theta)^2 + \xi = 0,$$

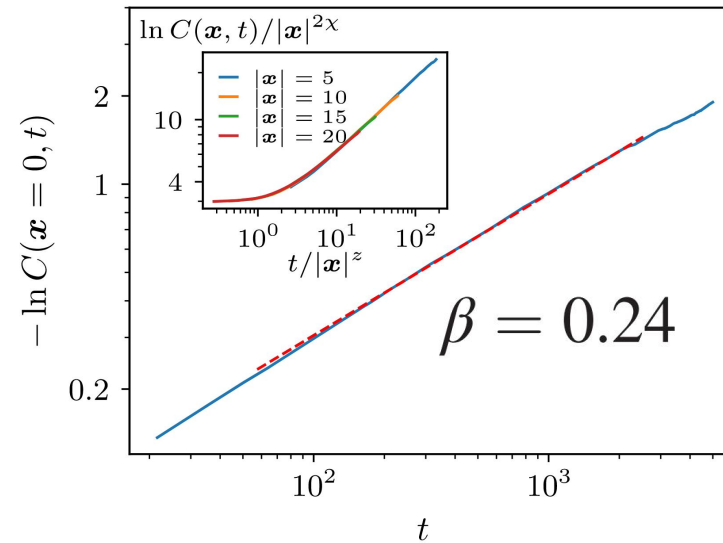
Chronons in the many-body Van der Pol system

$$\ln C(t, 0) \sim -A t^{2\beta}$$

$$\ln C(0, \mathbf{x}) \sim -B |\mathbf{x}|^{2\chi}$$



blue: high noise
red: low noise



2d kpz: $\beta \approx 0.24, \chi \approx 0.39$

Chronons in various systems

System	Type of CTC	Goldstone mode	KPZ type
Laser / polariton	Synchronized CTC	Phase fluctuation $\delta\theta$	KPZ
Kuramoto lattice	Synchronized CTC	Local phase field $\delta\theta(\mathbf{x}, t)$	KPZ
Magnon condensate	Traveling-wave CTC	Combined phase $\delta\theta(t, \mathbf{x})$	Anisotropic KPZ
Active XY model	Self-rotating CTC	Orientation phase $\delta\theta$	KPZ

conclusion and outlook

- **conclusion:** the goldstone mode “chronon” of continuous time crystal belong to kpz university class
- **outlook:** additional conserved quantities or the couplings to other soft modes may give rise to new universal behaviors.